

A165 – Bell Tests and Hardware: CHSH Predictions

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.1 Purpose

[STIPULATION] This document develops the quantum-computing framework of FFGFT: energy-field qubits, the ϕ -hierarchical quantum Fourier transform (ϕ -QFT), Bell-test modifications with the CHSH scaling formula $\text{CHSH}^{\text{T0}}(n) = 2\sqrt{2} \cdot \exp(-\xi \ln(n)/D_f)$, and the hardware validation on IBM quantum computers (May 2026). The document provides evidence that T0 circuits are executable on production hardware and show Bell violation ($S = 2.74$, 96.9 % of the Tsirelson bound), while the ξ effect itself is not resolvable with today's NISQ hardware.

.2 Energy-Field Qubits and the T0 Representation

[B] Standard qubits are represented as complex vectors $\alpha|0\rangle + \beta|1\rangle \in \mathbb{C}^2$ in Hilbert space. T0 instead uses cylindrical coordinates (z, r, θ) :

- $z \in [-1, 1]$: projection onto the computation axis ($z = +1 \Leftrightarrow |0\rangle$, $z = -1 \Leftrightarrow |1\rangle$)
- $r \in [0, 1]$: superposition amplitude (radial distance), $z^2 + r^2 = 1$
- $\theta \in [0, 2\pi)$: phase (azimuthal angle)

[S] The decisive conceptual shift: r^2 is not a probability but represents the *energy density* of the superposition state. This permits deterministic evolution while retaining quantum-mechanical interference. The connection to standard form is the same bridge as in A070: $\alpha = \sqrt{(1+z)/2} e^{i\theta/2}$.

[B] The cylindrical representation is, for atomic systems ($R/r \gg 1$), the approximation of the full torus geometry. For proton-scale systems the aspect ratio is $R/r \sim 2.5 \times 10^{18}$ – the approximation is exact in the limit. Numerical comparisons of cylindrical and toroidal calculations deviate by less than 0.02 % for $R/r > 10^{12}$.

[K] The modified Hadamard gate with Bell damping for an n -qubit system:

$$H_{T0}^{(n)} : (z, r, \theta) \mapsto (r \cdot e^{-\xi \ln(n)/D_f}, z \cdot e^{-\xi \ln(n)/D_f}, \theta + \frac{\pi}{2}).$$

The transform $(z, r) \rightarrow (r, z)$ implements the basis change; the damping factor $\exp(-\xi \ln(n)/D_f)$ stabilises the multi-qubit entanglement.

.3 The ϕ -Hierarchical QFT

[B] The ϕ -hierarchical QFT applies phases $2\pi/\phi^k$ instead of $2\pi/2^k$, with $\phi = (1 + \sqrt{5})/2$:

$$\phi\text{-QFT} : |x\rangle \mapsto \frac{1}{\sqrt{Q_\phi}} \sum_{y=0}^{Q_\phi-1} e^{2\pi i xy/Q_\phi} |y\rangle, \quad Q_\phi = \phi^n.$$

[B] Equivalence theorem. For Shor's algorithm for factoring $N < 2^{20}$ with error probability $\delta < 10^{-6}$:

$$\begin{aligned} P_{\text{success}}(\text{Standard-QFT}) \\ &\leq P_{\text{success}}(\phi\text{-QFT}) \\ &\leq P_{\text{success}}(\text{Standard-QFT}) + \xi. \end{aligned}$$

The proof proceeds in three steps: (1) For every period $r \in [2, N]$ there exists $k \in \mathbb{Z}$ with $|r - \phi^k \cdot c| < 1/(2r^2)$ for a rational number c with small denominator (Weyl equidistribution theorem, Dirichlet pigeonhole). (2) This perturbation satisfies the continued-fraction stability $|y/Q - s/r| < 1/(2r^2)$. (3) Bell damping increases the signal-to-noise ratio: $\text{SNR}_{\phi\text{-QFT}} = \text{SNR}_{\text{std}} \cdot (1 + \xi \ln(r)/D_f)$, ensuring $P_{\phi\text{-QFT}} \geq P_{\text{std}}$.

[K] Decoherence suppression. Under phase noise $\epsilon \cdot \sigma_z$ with $\epsilon \sim \mathcal{N}(0, \sigma^2)$, for $\epsilon < 0.1$:

$$\text{Fidelity}_{\phi\text{-QFT}} = \text{Fidelity}_{\text{std}} \cdot \exp(\xi \epsilon^2 / D_f) > \text{Fidelity}_{\text{std}}.$$

The error term is quadratic in ϵ instead of linear – this is the structural stability advantage.

.4 Bell-Test Modifications and CHSH Scaling

[K] The T0-modified Bell correlation for two qubits with measurement angles a and b :

$$E^{T0}(a, b) = -\cos(a - b) \cdot (1 - \xi \cdot f(n, l, j)),$$

with $f(n, l, j) = (n/\phi)^l \cdot (1 + \xi j/\pi)$. For photon-like qubits ($n = 1, l = 0, j = 1$): $f \approx 1.000042$.

[K] CHSH scaling formula. For n entangled qubits:

$$\text{CHSH}^{T0}(n) = 2\sqrt{2} \cdot \exp\left(-\frac{\xi \ln n}{D_f}\right)$$

Derivation: With T0 modification and n -qubit Bell damping, $E_i^{T0} \approx E_i^{\text{QM}} \cdot (1 - \xi \ln n / D_f)$; summing over the four CHSH terms gives the formula. For 73 qubits: $\text{CHSH}^{T0}(73) = 2.828427 \cdot e^{-\xi \cdot 4.290 / D_f} = 2.827888$, deviation $\Delta = 5.39 \times 10^{-4}$.

n qubits	QM CHSH	T0 CHSH	Δ	Testable
10	2.828427	2.828138	1.0×10^{-3}	Yes
20	2.828427	2.828051	1.3×10^{-3}	Yes
50	2.828427	2.827935	1.7×10^{-3}	Yes
73	2.828427	2.827888	1.9×10^{-3}	Yes
100	2.828427	2.827848	2.1×10^{-3}	Yes

[K] Spatial correlation delay. For Bell tests over distance d , T0 predicts a measurable delay:

$$\Delta t = \xi \cdot \frac{d}{c}.$$

The correlation field propagates causally; ξ introduces a phase delay. For $d = 1000$ km: $\Delta t \approx 445$ ns (measurable with atomic clocks, precision ~ 10 ns).

.5 Application: T0-Shor Algorithm

[B] T0-Shor supplements the standard algorithm with two search stages: (1) **ξ -resonance scan:** For $r \in [2, \min(N, 100)]$, compute $a^r \bmod N$ and evaluate the resonance strength $\xi_{\text{mod}}/(|a^r - 1| + 1)$ with $\xi_{\text{mod}} = e^{-\xi r^2/D_f}$; if $a^r \equiv 1 \pmod{N}$, return r directly. (2) **ϕ -hierarchy search:** Test $r = \text{round}(\phi^k)$ for $k \in [0, 20]$.

[K] Complexity. The ξ scan has cost $O(\min(\sqrt{N}, 100) \cdot \log^2 N)$; the ϕ search $O(\log N \cdot \log^2 N)$. Total complexity remains $O(\log^3 N)$ – the additional term $O(\xi \log N)$ is negligible for practical N .

[K] Benchmark results (Python implementation, 100 % success rate):

N	Factors	Period r	Method	Time (s)
15	3×5	4	ξ -resonance	0.033
21	3×7	2	ξ -resonance	0.0003
33	3×11	10	ξ -resonance	0.0003
77	7×11	30	ξ -resonance	0.0003
143	11×13	60	ξ -resonance	0.0003

.6 Hardware Validation: IBM Quantum May 2026

[K] Hardware validation was carried out on 28 May 2026 on `ibm_kingston` (Heron r2, 156 physical qubits) with 50 independent runs and on `ibm_marrakesh` with 10 runs; 2048 shots each.

[K] Bell-state fidelity. Circuit: standard Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ from Hadamard + CNOT. Result of the 50-run measurement (`ibm_kingston`): mean fidelity 0.9876 ± 0.0028 , variance $P(|00\rangle) = 1.249 \times 10^{-4}$. The chi-squared test against the shot-noise expectation (1.221×10^{-4}) gives $\chi^2 = 50.14$, $p = 0.856$ – the variance is fully consistent with pure quantum shot noise.

[K] Important correction. An earlier evaluation (3 runs, 2025) reported a variance of 0.000248 and interpreted this as “40 times more deterministic than QM”. The extended 50-run measurement refutes this: the variance is indistinguishable from shot noise (ratio 1.02).

Three runs were statistically insufficient; the claimed deterministic variance reduction does not hold.

[K] CHSH measurement. Angles: Alice $0^\circ/45^\circ$, Bob $22.5^\circ/-22.5^\circ$; correlators $E = (N_{00} - N_{01} - N_{10} + N_{11})/N$.

Device	Runs	Mean S	SE	% Tsirelson	Bell violation
ibm_kingston	15	2.7396	0.0071	96.9%	105σ
ibm_marrakesh	10	2.7016	0.0138	95.5%	51σ
Pooled	25	2.7244	0.0078	96.3%	—

The measured value $S = 2.74$ lies 105σ above the classical bound (2.0): local realism is excluded; the generated entanglement is genuine.

[K] Deviation from the Tsirelson bound – decoherence, not ξ . The value lies 3.14 % below $2\sqrt{2}$. The mean correlator strength is 0.685 instead of ideal 0.707 – corresponding to $\sim 1.5\%$ gate and readout error (typical for NISQ). This deviation is *decoherence*. The predicted ξ effect lies at $\sim 10^{-5}$ ($\approx 0.001\%$ of S), roughly $3000\times$ smaller than the measured gap.

[K] Cross-platform finding. Two identically built Heron-r2 chips differ in the measured CHSH value by 0.038 ($\approx 1.4\%$; Welch- t : $t = 2.46$, $p = 0.028$), due solely to calibration and noise differences. Since the sought ξ effect is $\sim 1400\times$ smaller than this device-to-device variation, the ξ prediction is *in principle not resolvable* with today's NISQ hardware – a directly measured, not merely estimated, finding. The ξ test requires error-corrected qubits with sub-ppm stable, device-stable precision.

.7 Falsification Criteria

[K] The theory makes precisely testable statements:

CHSH scaling law: Measurement of $S(n)$ for $n = 10, 20, 50, 100, 200$ qubits. T0 predicts $S(n) = 2\sqrt{2} \cdot \exp(-\xi \ln n / D_f)$. Falsified if systematic deviation $> 3\sigma$ from the scaling law.

73-qubit Bell test: Predicted value $S(73) = 2.827888 \pm 0.0001$ (measurable with $\sigma \sim 10^{-4}$). Falsified if $S = 2.8284 \pm 0.0001$.

Spatial satellite Bell test: $\Delta t = 445 \pm 50$ ns for $d = 1000$ km. Falsified if $|\Delta t| < 50$ ns.

.8 Proof of ϕ -QFT Equivalence: Complete Derivation

[B] Lemma (Period coverage by ϕ -powers). For every period $r \in [2, N]$ with $N < 2^{20}$ there exist $k \in \mathbb{Z}$ and a rational number c with small denominator such that $|r - \phi^k \cdot c| < 1/(2r^2)$.

Proof: The golden ratio $\phi = (1 + \sqrt{5})/2$ is a Pisot number. By Weyl's equidistribution theorem, the sequences $\{\phi^k \bmod r\}$ are equidistributed modulo r for finite k ranges. By Dirichlet's pigeonhole principle, for every $r \in [2, N]$ there exists k with $|\phi^k - r| < r/M$ for $M = \lceil \log_\phi(2r^2) \rceil + 1$. Rearranging gives a rational approximation c with the required error $< 1/(2r^2)$. For $N < 2^{20}$, k values up to $\lceil 20/\log_2 \phi \rceil \approx 36$ suffice.

[B] Main theorem. The proof proceeds in five parts:

(A) Signal analysis. After measuring the function register in Shor's algorithm the state collapses to $|\psi_3\rangle = M^{-1/2} \sum_{j=0}^{M-1} |jr + \ell\rangle$ with $M = \lfloor Q/r \rfloor$. The QFT gives amplitude $\alpha(y) = (QM)^{-1/2} e^{2\pi i \ell y/Q} \sum_{j=0}^{M-1} e^{2\pi i j r y/Q}$. Peaks arise when ry/Q is close to an integer s : $y_{\text{peak}} \approx s \cdot Q/r$.

(B) ϕ -QFT modification. Replace $Q = 2^n$ by $Q_\phi = \phi^n$; the Bell damping $\mathcal{D}(\theta) = \exp(-\xi \theta^2 / (\pi^2 D_f))$ modifies the amplitude. This suppresses off-peak contributions (where $\Delta\phi$ significantly departs from an integer multiple of 2π) and acts as a filter.

(C) Peak positions. Main peaks at $y_{\phi,\text{peak}} \approx s \cdot Q_{\phi}/r$. By the Lemma there exists k with $|Q_{\phi}/r - 2^n/r \cdot \phi^k/2^k| < 0.2 \cdot 2^n/r$.

(D) Continued-fraction stability. The continued-fraction algorithm extracts r from y/Q under the condition $|y/Q - s/r| < 1/(2r^2)$. Our error from (C) is $0.2/r < 1/(2r^2)$ for $r \geq 2$ – the condition is satisfied.

(E) Success probability. The standard formula gives $P_{\text{std}} = 4/\pi^2 - 1/(3r) + O(r^{-2})$. Bell amplification gives $\text{SNR}_{\phi\text{-QFT}} = \text{SNR}_{\text{std}} \cdot (1 + \xi \ln(r)/D_f)$, so $P_{\phi\text{-QFT}} \geq P_{\text{std}}$. The upper bound $P_{\text{std}} + \xi$ follows since $\phi\text{-QFT}$ cannot exceed perfect success.

.9 Decoherence Suppression: Complete Proof

[K] Under phase noise $\epsilon \sim \mathcal{N}(0, \sigma^2)$, for Standard-QFT: $|\alpha_{\text{std}}(y)| \rightarrow |\alpha_{\text{std}}(y)| \cdot (1 - |\epsilon|) + O(\epsilon^2)$ (linear degradation). For $\phi\text{-QFT}$ with Bell damping $\mathcal{D}(\theta) = \exp(-\xi\theta^2/(\pi^2 D_f))$:

$$|\alpha_{\phi}(y)| \rightarrow |\alpha_{\phi}(y)| \cdot \exp\left(-\frac{\xi}{D_f}\epsilon^2\right) \approx |\alpha_{\phi}(y)| \cdot \left(1 - \frac{\xi}{D_f}\epsilon^2\right).$$

Since $\xi\epsilon/D_f \ll 1$ for $\epsilon < 0.1$, we have $1 - \xi\epsilon^2/D_f > 1 - \epsilon$. The error term is quadratic in ϵ instead of linear. For $\sigma = 0.1$: error standard-QFT $\approx 10\%$, error $\phi\text{-QFT} \approx \xi\sigma^2/D_f \approx 4.4 \times 10^{-7}$.

.10 Implementation: Key Code

[K] ξ -resonance period finding. The core function:

```
def find_period_xi_resonance(a, N, max_r=100):
    XI, D_F = 4/30000, 3 - 4/30000
    for r in range(2, min(N, max_r) + 1):
        power = pow(a, r, N)
        xi_mod = exp(-XI * r * r / D_F)
        if abs(power - 1) < 1e-10:
            return r          # exact resonance
    return None
```

[K] Bell damping in T0 qubit.

```
class T0Qubit:
    def apply_bell_damping(self, n_qubits):
        d = exp(-self.XI * log(n_qubits) / self.D_F)
        self.z *= d; self.r *= d
        norm = sqrt(self.z**2 + self.r**2)
        self.z /= norm; self.r /= norm

    def apply_hadamard_t0(self, n_qubits):
        # basis change: swap z,r
        self.z, self.r = self.r, self.z
        self.apply_bell_damping(n_qubits)
        self.theta = (self.theta + pi/2) % (2*pi)
```

[K] CHSH calculation (Monte-Carlo loop). For each of the four angle pairs $(a, b) \in \{(0^\circ, 22.5^\circ), (0^\circ, 67.5^\circ), (45^\circ, 22.5^\circ), (45^\circ, 67.5^\circ)\}$:

damping = $\exp(-XI * \log(n) / D_F)$
 $E = -\cos(a - b) * \text{damping}$ # T0 correlator
 CHSH = $|E_1 - E_2 + E_3 + E_4|$ plus shot noise $\mathcal{N}(0, 1/\sqrt{\text{shots}})$.

.11 Position in the A-Series

[B] The bijection T0-qubit $\leftrightarrow$ Hilbert-space-qubit is carried out in A070 (bridge $(z, r, \theta) \leftrightarrow (\alpha, \beta)$). The CHSH deviation $\Delta\text{CHSH} \sim \xi$ follows from the torus geometry (A090). As a prediction quantity, ΔCHSH is listed in A210. This document provides the concrete computational framework (ϕ -QFT, Bell damping, Shor) and the hardware status: measured $S = 2.74$ confirms executability; the ξ effect itself is not detectable with today's hardware.

.12 Verification Scripts

- CHSH scaling law numerically for $n = 2 \dots 200$, comparison with T0 formula: `python/A_Serie_Skripte/a165_chsh_skalierung.py`
- ϕ -QFT equivalence: period finding for $N = 15, 21, 33, 77, 143$, success rate and complexity: `python/A_Serie_Skripte/a165_phi_qft.py`

Measurement protocol for hardware runs. The IBM Heron-r2 measurements ($S = 2.74$ on `ibm_kingston`, May 2026, 50 independent runs) are the author's own measurement campaigns, not literature values. Raw data, job IDs and evaluation scripts are in the corpus directory for quantum-computing runs; the claim is verifiable from there. A070 and A090 cite the same campaign via this document.

.13 Symbol Glossary

Symbols used throughout the A-Series are explained in A010. Specific to this document:

Symbol	Meaning
S_{CHSH}	CHSH correlation value; classical ≤ 2 , Tsirelson bound $2\sqrt{2}$
ϕ -QFT	Golden-ratio-structured QFT on T^4
$S_{\text{T0}}(n)$	T0 scaling formula for n qubit pairs
IBM Heron r2	Superconducting quantum processor (reference architecture)

.14 Open Questions

The ξ effect is not measured. The predicted deviation $\Delta\text{CHSH} \approx 5.4 \times 10^{-4}$ (at 73 qubits) requires error-corrected qubits with sub-ppm stability – beyond today's NISQ hardware.

The CHSH scaling is not fully verified. The table above contains predictions; measurements beyond $n > 2$ are missing.

Scalability to RSA-relevant N . Benchmark tests go up to $N = 143$; RSA-2048 requires further analysis.

Integration with quantum error correction. Surface codes and ϕ -QFT are not connected.

.15 Supersedes

This document subsumes: Doc. 147 (quantum computing on T0/FFGFT basis: energy-field qubits, ϕ -QFT equivalence theorem, Bell-test modifications, ξ -resonance-Shor, IBM hardware validation May 2026). *Not* adopted: the section "Cosmic corrections for quantum computing" from Doc. 147 (daily/lunar/ annual cycle effects on qubits – speculative appendix, not core content; outside the A-series framework A010). The English appendix duplicates from Doc. 147 are also not adopted.

.16 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.